

POC 06: Enterprise Automation Platform - IaC, Jenkins And Monitoring

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Role: **AWS DevOps Engineer**

Public-safe name: **Enterprise Automation Platform**

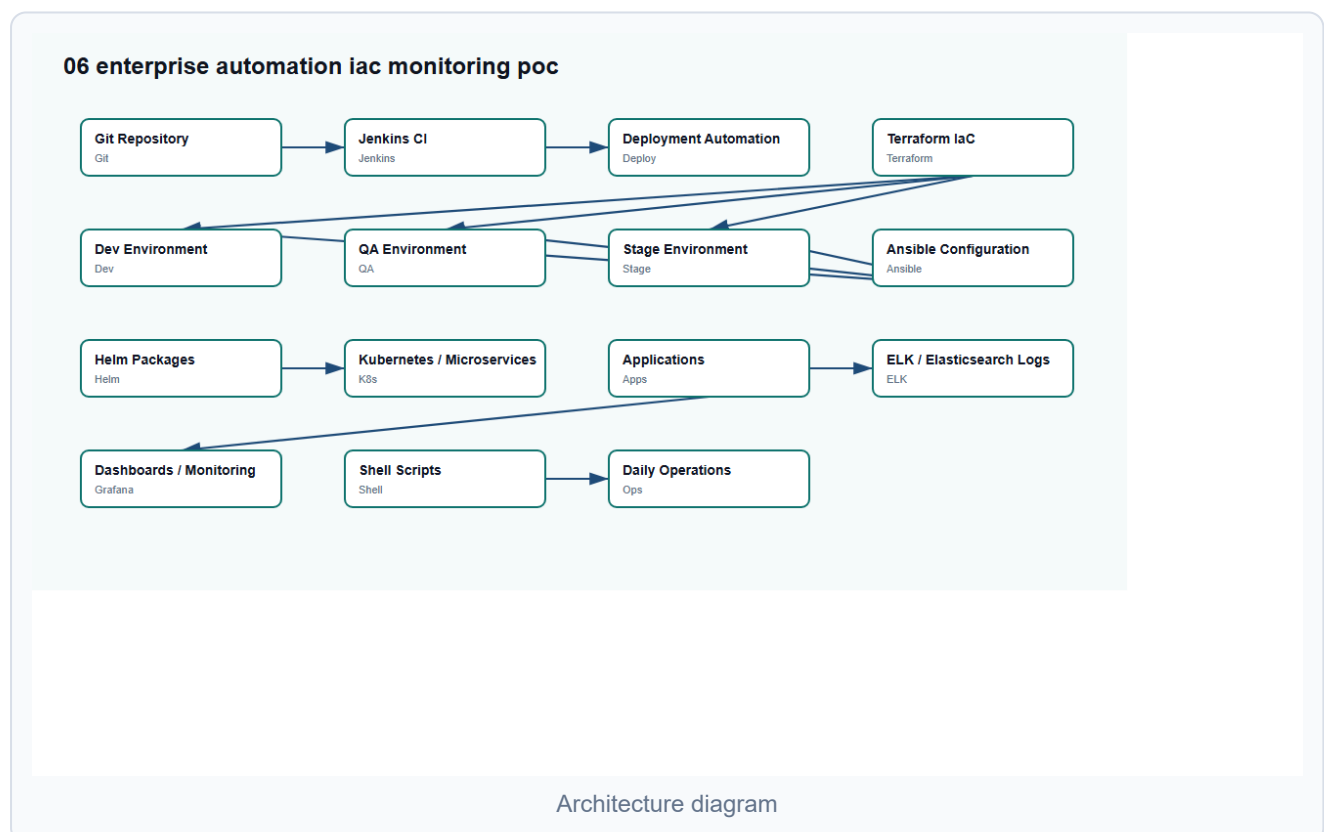
Confidentiality Note

This POC excludes internal environment values, repository URLs, server IPs, credentials, dashboard links, and client-specific implementation details.

POC Objective

Create a repeatable infrastructure and operations blueprint using Terraform, Ansible, Jenkins, Helm, Kubernetes, ELK/Elasticsearch, dashboards, and shell scripting for Dev, QA, and Stage environments.

IaC And Operations Diagram



What I Worked On

- â€¢ Automated infrastructure provisioning for Dev, QA, and Stage environments using Terraform.
- â€¢ Used Ansible for configuration management, application configuration, package management, and operational commands.
- â€¢ Installed, configured, and administered Jenkins CI on Linux machines.
- â€¢ Created dashboards and management reports using ELK/Elasticsearch.
- â€¢ Developed Helm packages for microservice deployment and log management.
- â€¢ Automated recurring operational tasks with shell scripts.

POC Implementation Scope

- â€¢ Terraform module skeleton for repeated environments.
- â€¢ Ansible playbooks for package/configuration setup.
- â€¢ Jenkins pipeline for build and deployment.
- â€¢ Helm chart structure for dummy microservice.
- â€¢ ELK/Grafana monitoring documentation.
- â€¢ Shell scripts for operational automation.

Suggested Repository Structure

```
iac-jenkins-monitoring-poc/  
|-- README.md  
|-- terraform/  
|   |-- modules/  
|   `-- envs/  
|-- ansible/  
|   |-- inventory.ini  
|   `-- playbook.yml  
|-- helm/  
|   `-- dummy-service/  
|-- jenkins/  
|   `-- Jenkinsfile  
|-- scripts/  
|   `-- daily-checks.sh  
`-- docs/  
    |-- monitoring.md  
    `-- runbook.md
```

Validation Steps

Run Terraform `fmt` and `validate`.

Run Ansible syntax check.

Review Jenkins pipeline stages.

Render Helm templates.

Run shell script lint/checks where available.

Expected Outcome

â€¢ Repeatable non-production infrastructure pattern.

â€¢ Better CI/CD and configuration management.

â€¢ Improved logging, dashboards, and operational automation.

Technologies

Terraform, Ansible, Jenkins, Linux, AWS, Git, GitHub, Helm, Kubernetes, ELK/Elasticsearch, Grafana, Shell scripting, CI/CD.